Teaching Statement: Undergraduate Electrical and Computer Engineering

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Connecting engineering fundamentals to implementation

My objective as an ECE instructor is to help students understand a mathematical model, implement or measure its behavior, and explain the differences between predicted and observed results. Students should learn to justify design choices, test assumptions and communicate an engineering conclusion rather than simply obtain a numerical answer or a working program.

At the University of Lahore, I taught undergraduate Computer Programming, Communication Systems, and Wireless Communication. I also supported communications laboratories, instructional materials and assessment at UET Taxila and mentored research students at COMSATS University Islamabad and the University of Oklahoma. These experiences reinforced the importance of clear prerequisites, worked examples, hands-on practice and timely feedback. My industry and doctoral work later strengthened my emphasis on reproducibility, realistic measurements, and disciplined validation.

Teaching interests and laboratory activities

My strongest teaching interests are Computer Programming, Communication Systems, Wireless Communications, Signals and Systems, Digital Signal Processing, computer networking, and applied AI/ML for ECE. I would also contribute to engineering computation, data analysis and image processing, and consider introductory digital systems or control systems after targeted refresher study and review of Purdue's course and laboratory requirements.

I would organize instruction around explicit outcomes and progressively more independent tasks. In programming, students could trace an algorithm, implement it in Python or C/C++, test edge cases, and explain its limitations. In signals and systems, students could derive the sampling relationship, simulate aliasing, and verify the result through a waveform or frequency-domain plot. In communications, a laboratory could compare simulated bit-error rates at different signal-to-noise ratios, evaluate the effect of channel assumptions, and document measurement uncertainty. These are prospective activities, informed by my experience, not claims that I previously taught those exact courses or laboratories.

Applied ML provides another bridge between computation and engineering. Students might compare an interpretable baseline with a neural network on open sensor or network data and evaluate generalization, false positives, and resource requirements. Purdue Indianapolis's interests in autonomous systems and intelligent software offer possible contexts for examples involving sensing, communications and reliable computational methods. Projects would use approved equipment and open or appropriately authorized data rather than proprietary operational records.

Assessment, course coordination and teaching-assistant supervision

I understand the expectation of three undergraduate lecture and/or laboratory courses each semester and primary responsibility for their teaching and assessment. I would plan each semester using a shared course calendar, learning outcomes, laboratory readiness checks, assessment milestones and regular office hours. Common rubrics would distinguish mathematical reasoning, implementation, experimentation, documentation and interpretation. Short formative exercises and review sessions would reveal misconceptions before higher-stakes examinations.

For assigned graduate and undergraduate teaching assistants, I would establish clear responsibilities, review laboratory procedures together, provide sample grading solutions and calibrate rubrics using common examples. Brief regular coordination meetings would address student questions, grading consistency, safety concerns and upcoming teaching tasks. I would retain responsibility for course decisions and final assessment. My prior experience includes student mentoring and laboratory support; formal supervision of assigned Purdue TAs would be a responsibility I would undertake in accordance with school procedures.

Responsible use of AI in engineering education

I have practical experience with machine learning, generative AI tools, retrieval-augmented generation and agentic workflows through industry work, research and professional learning. In teaching, I would use appropriate AI tools to help develop examples, explore alternative explanations and support debugging where permitted. Students could compare an AI-generated implementation with an independently derived solution, identify incorrect assumptions and write tests to validate the output. AI-assisted work should be disclosed when required, and students should not enter confidential data into unauthorized tools.

I would preserve meaningful independent assessment through in-class reasoning, individual examinations, oral explanations, code walkthroughs and laboratory demonstrations. The goal is not to prohibit useful tools indiscriminately, but to make students responsible for the correctness and integrity of their work under Purdue’s policies.

Student mentoring and continuous improvement

Students enter ECE with different levels of mathematics, programming and laboratory experience. I would make expectations visible, offer accessible starter materials, use structured group roles and provide constructive feedback without lowering academic standards. For projects and capstones, I would guide teams through requirements, a baseline, design decisions, implementation, verification and communication, with individual accountability.

I would welcome participation in curriculum development and school educational service, using evidence from student work, feedback and discussions with colleagues to improve instruction. My aim is to prepare students who possess sound technical fundamentals, can collaborate and communicate effectively, and can evaluate engineering results with confidence and intellectual integrity.